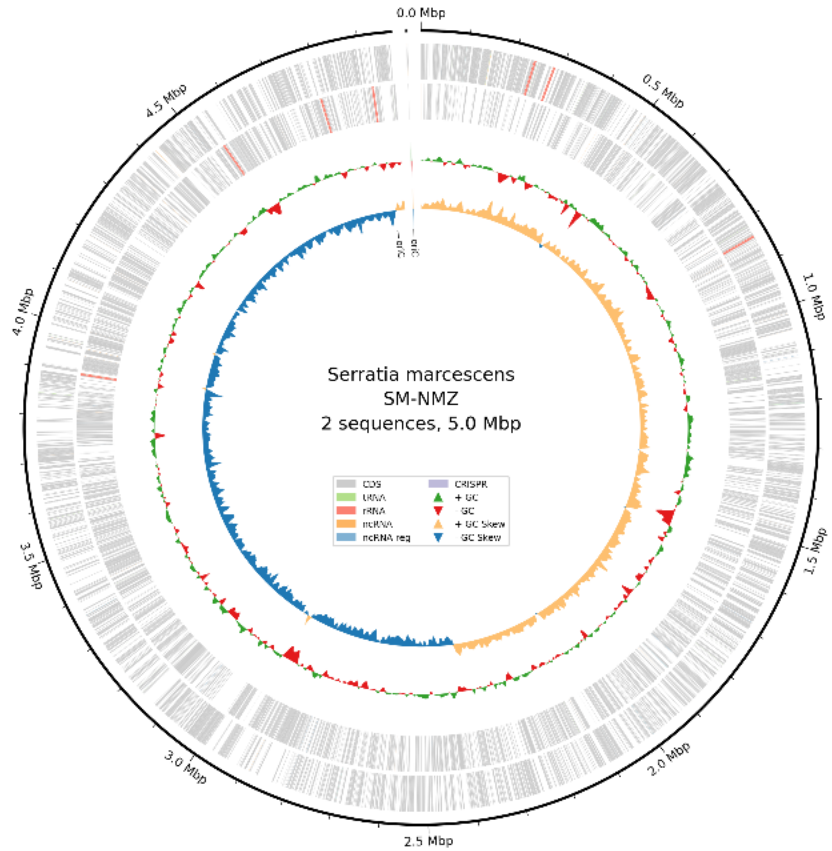
NMZ Serratia genomics

An explainer for the research group

SM-NMZ Slaughterhouse effluent
C14-NMZ Calotropis rhizosphere



20 October 2026

DNA, gene predictions and activity

Stage	What it supplies
Sequencing	DNA fragments called reads
Assembly	Reconstructed chromosome and other replicons
Annotation	Predicted genes and possible functions
Experiment	An observed effect under stated conditions

A predicted function is a candidate for explanation.

What each kind of evidence answers

Evidence	Question it addresses
Related gene or protein	Is there a plausible candidate?
Several linked genes	Does the proposed machinery fit together?
Chemical analysis	Which molecular assignment has support?
Activity assay	What effect occurred in this experiment?
Controlled perturbation	Does this factor contribute to that effect?

Genome resources and identity

Feature	SM-NMZ	C14-NMZ
Chromosome	4.971 Mb	5.051 Mb
Additional replicon	3.581-kb plasmid	None recovered
Bakta CDSs	4,566	4,626
GTDB r232 label	S. marcescens	S. nematodiphila

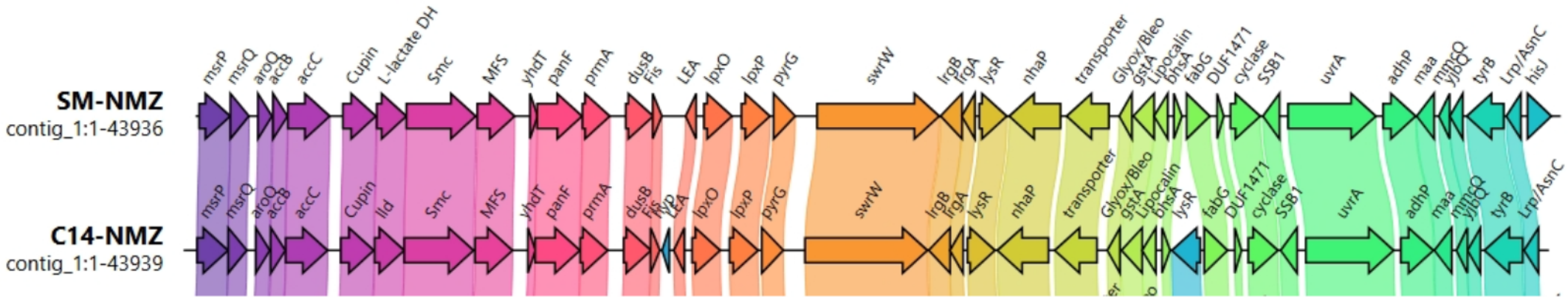
Unified TYGS: one species cluster, separate subspecies clusters

Final TYGS total lengths now match Bakta exactly.

SwrW and reported serratamolides

Genomic evidence

Both isolates carry a conserved swrW candidate.



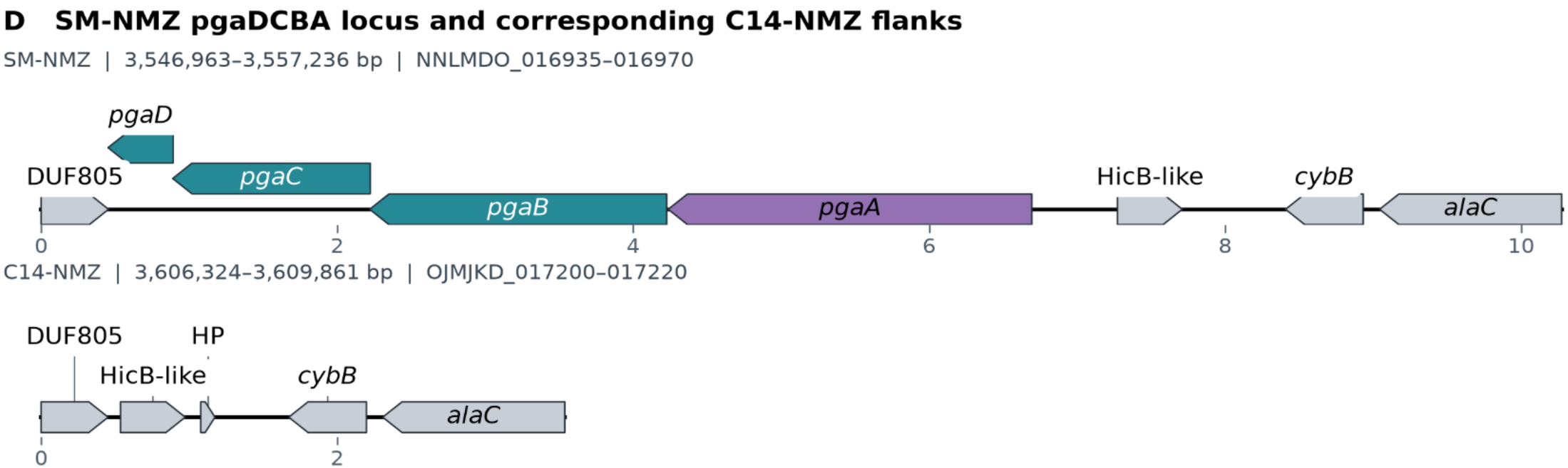
Reported chemical evidence

LC-HRMS: serrawettin W1 and serratamolide C

The pig locus also matches the red/pink culture phenotype.

Extracellular matrices and surface activity

Both isolates: capsule/EPS and cellulose candidates



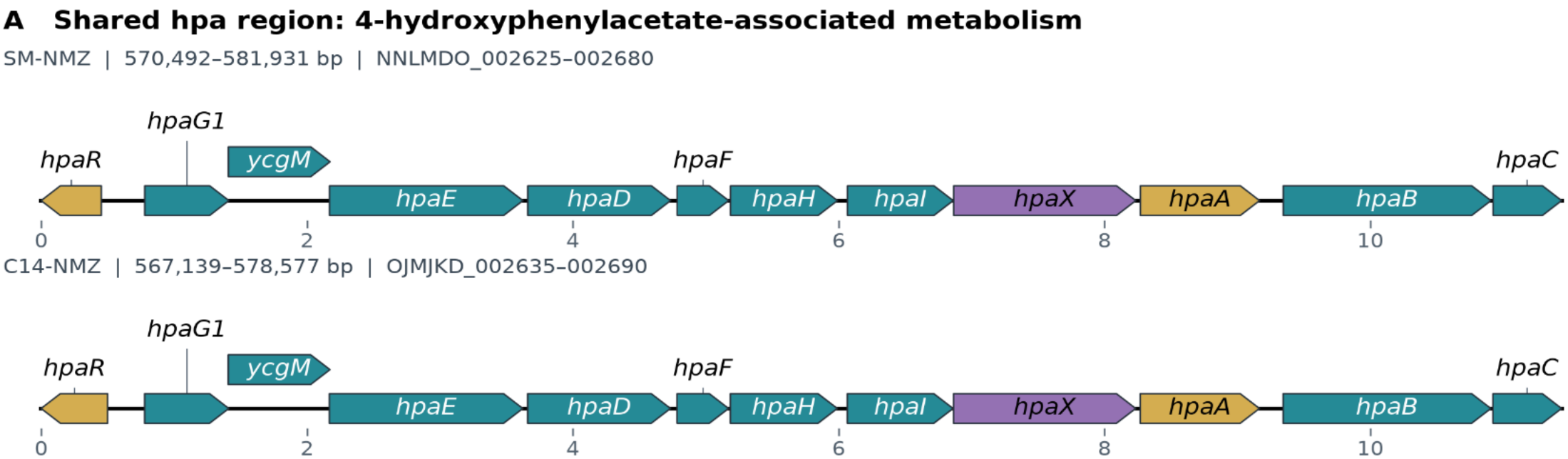
Gene arrows. Scale in kb from the first displayed CDS.

SM has the contiguous pga locus associated with PNAG.

Matrix contribution depends on the material actually assayed.

Aromatic metabolism depends on the substrate

Both isolates: organized hpa and paa loci



Gene arrows. Scale in kb from the first displayed CDS.

SM adds hydroxybenzoate/pca and gentisate candidates.

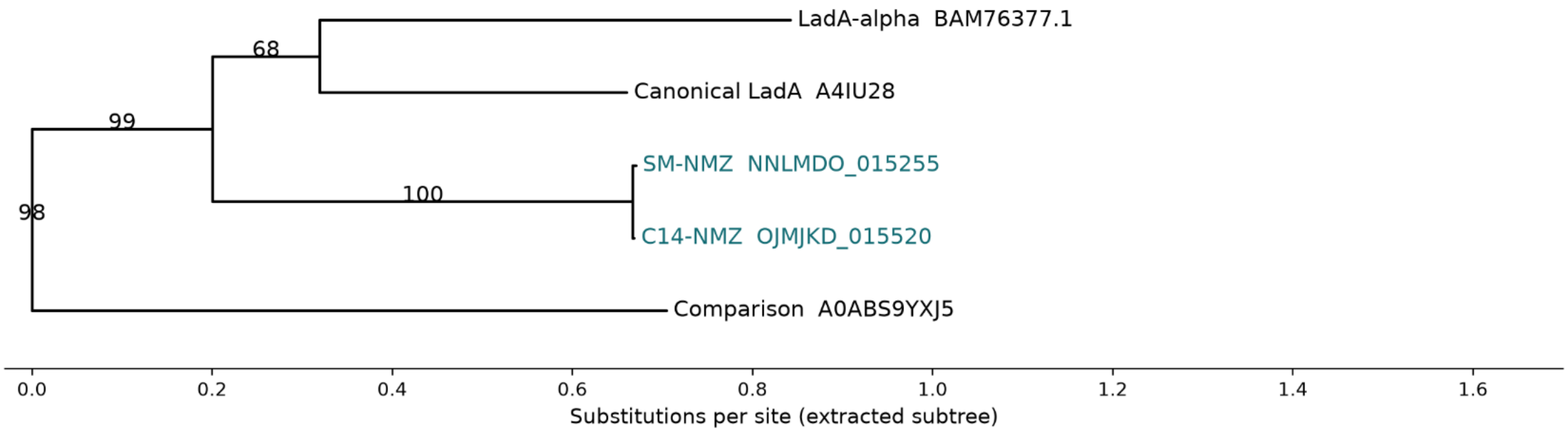
These findings concern specific oxygenated aromatics.

The first alkane oxidation remains unresolved

Simplified step	Genomic interpretation
Alkane to alcohol	Initiating system unresolved
Alcohol to aldehyde	Downstream processing candidates
Aldehyde to fatty acid	Downstream processing candidates
Fatty acid to cellular metabolism	Beta-oxidation and assimilation context

DNA-repair AlkB and alkane hydroxylase AlkB are different proteins.

The LadA-like comparison



Node labels show UFBoot support (%).

Strong sequence grouping supports a candidate relationship.

Its substrate remains unresolved. Nearby genes add sulfur/methionine context.

Matching assays to genomic candidates

Endpoint, if measured	Relevant interpretation	Evidence to pair with it
Surface/interfacial tension	Amphiphile contribution	Material, interface, mN/m, control
E24 emulsification index	Amphiphile and/or matrix contribution	Oil, 24-h protocol, layer heights
Aromatic conversion	Route matching the exact substrate	Substrate, products and recovery controls
Pour point	Low-temperature oil-flow performance	Oil, dose, method and temperature

Numerical endpoint data remain with the experimental team.

Host-interaction genes and application context

Reference-associated system	Biological role in characterized systems
ShlA–ShlB	Hemolysin machinery
Has	Heme acquisition
Type VI secretion system	Bacterial competition

Reference matches do not quantify pathogenicity or establish safety.

The genomics contribution to the paper

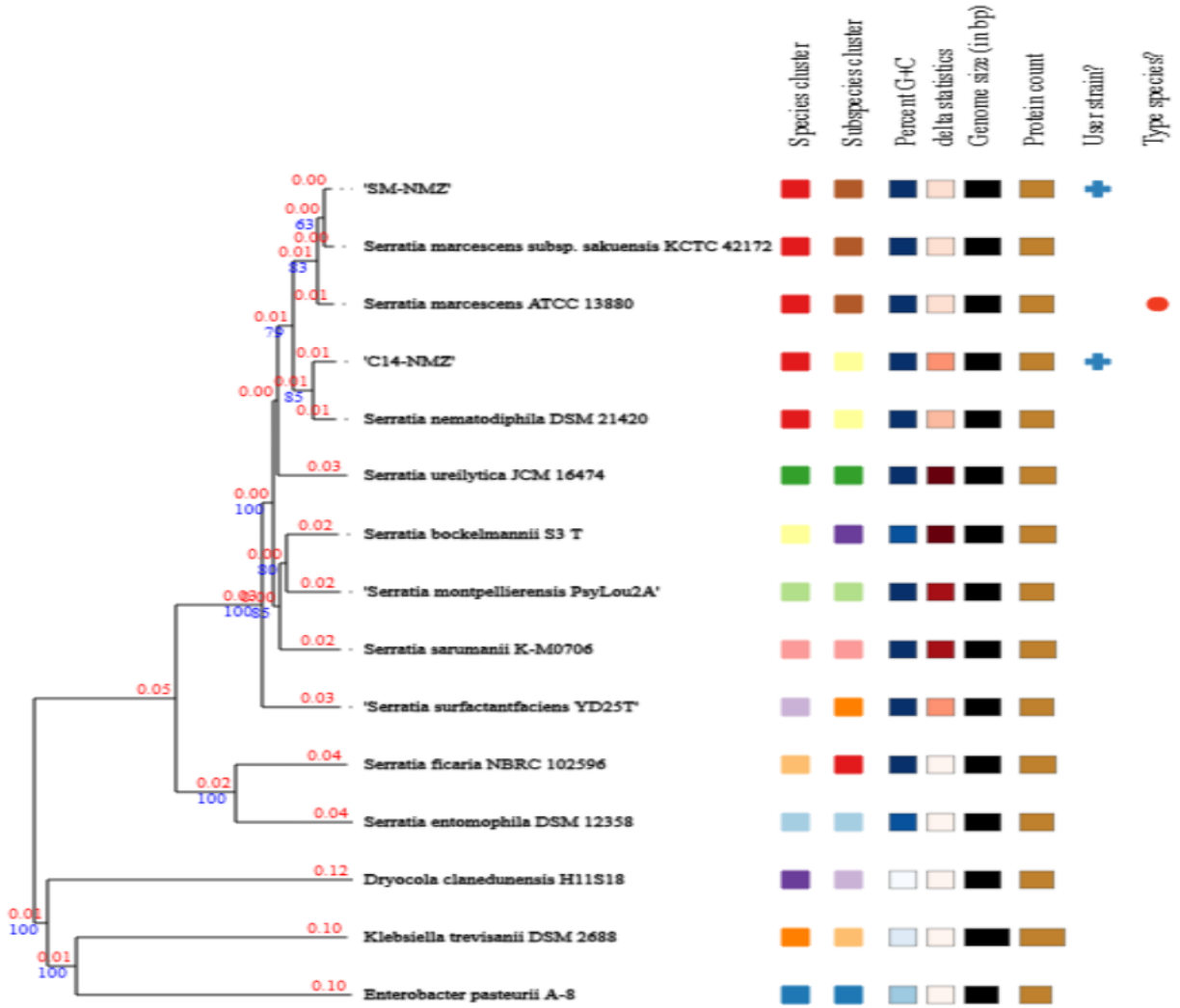
Finding	Contribution
SwrW and reported serratamolides	Specific surface-active metabolite lead
pig locus and red/pink cultures	Pigment-associated finding
Matrix loci and SM pga cluster	Shared systems plus a focal difference
Aromatic loci and LadA-like comparison	Substrate-specific context and an unresolved enzyme

Which measured endpoint and material should lead the paper?

Workflow and tools by purpose

Question	Tools and evidence
Which read evidence enters assembly?	NanoPlot, Chopper, Filtlong
How are sequences reconstructed?	Autocycler/assemblers, dnaapler, Dorado
Where are genes and what are candidates?	Bakta, antiSMASH, DRAM, CANT-HYD, HADEG
How are genomes and proteins related?	TYGS, GTDB-Tk, ProteinOrtho/PoFF, IQ-TREE
How are local arrangements compared?	clinker and DNA Features Viewer

The unified TYGS genome phylogram



15 genomes
13 type-strain references
2 focal isolates

One species cluster
Two focal subspecies clusters

GBDP d5 distances
100 pseudo-bootstrap replicates

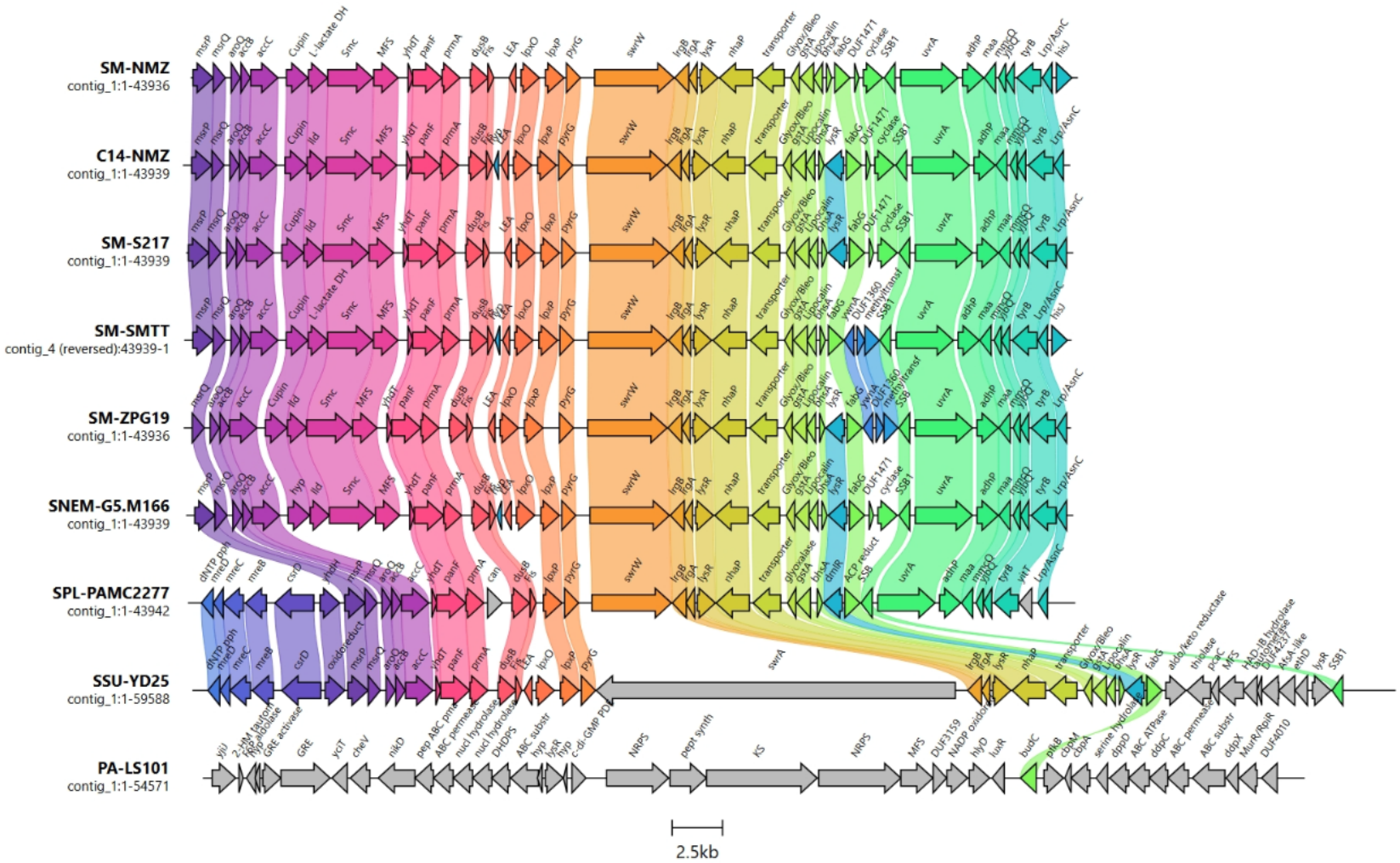
Taxonomy comparisons and length parity

dDDH(d4), %	SM-NMZ	C14-NMZ
S. marcescens ATCC 13880	90.7	73.7
S. nematodiphila DSM 21420	74.7	85.5
Direct focal comparison	74.2 (71.2–77.0)	Same comparison

TYGS total = Bakta total

SM 4,970,517 + 3,581 = 4,974,098 bp
C14 5,050,553 bp

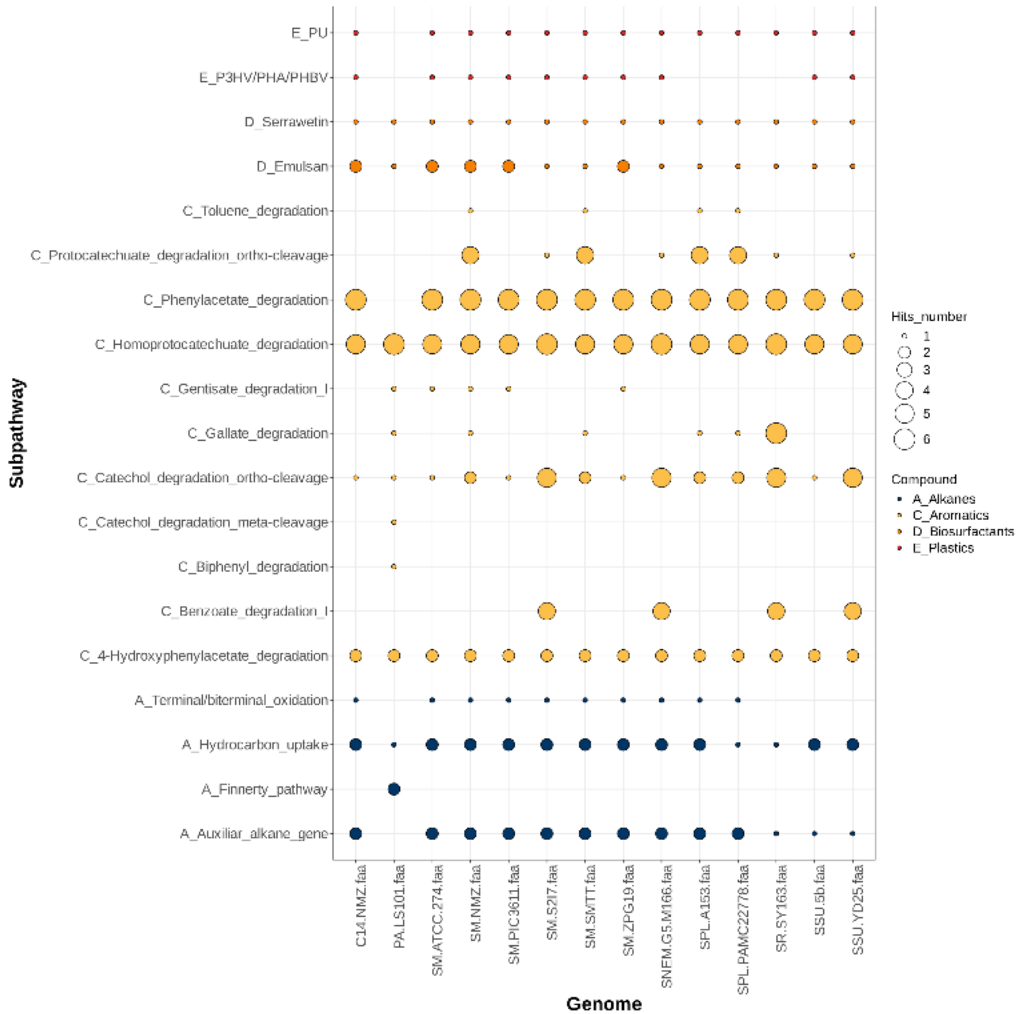
Reading the SwrW neighborhood



Arrows
Gene length and direction

Ribbons
Protein matches

Reading HADEG marker categories



Emulsan
Generic capsule markers

Toluene_degradation
Hydroxybenzoate-hydroxylase candidate

Bubble counts describe markers.
They do not measure pathway activity.

Chemical assignments and peak areas

Observation	What it can support
Exact mass and adduct	A candidate molecular composition
Fragmentation and reference match	Additional structural evidence
Authentic standard, when documented	A stronger identity comparison
Relative peak area	Detector signal under a stated denominator

The source reports chemical assignments. Genomics adds biosynthetic context.

Additional SM aromatic candidates

C SM-NMZ hydroxybenzoate-hydroxylase / pcaK neighborhood

SM-NMZ | 2,461,766–2,467,540 bp | NNLMDO_011760–011775



D SM-NMZ protocatechuate-associated region

SM-NMZ | 3,121,817–3,130,237 bp | NNLMDO_014965–015005

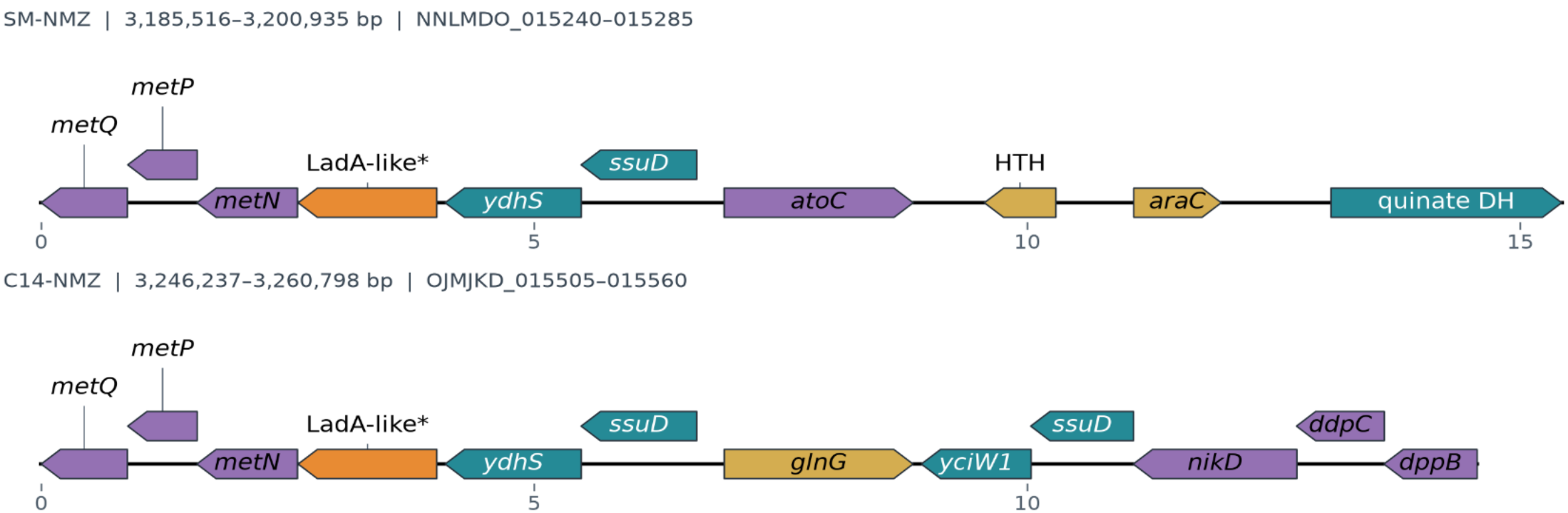


Scale in kb within each panel. Panels use different physical scales.

Additional candidates broaden the substrate hypotheses.

Some downstream steps remain unresolved.

The focal LadA-like genomic neighborhoods



Gene arrows. Scale in kb from the first displayed CDS.

Source gene qualifier: ssuD. Candidate substrate remains unresolved.